

Bolla Lokesh Reddy

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EDUCATION

Indian Institute of Technology Hyderabad

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 8.3/10

EXPERIENCE

Adobe – Content Intelligence Team

May 2026 – Jul 2026

Machine Learning Engineering Intern

India

- Developed an end-to-end email layout featurization pipeline combining DOM geometry with multimodal semantic representations for structural analysis.
- Designed deterministic extraction of 22 structural features and evaluated DINOv2/Gemini embeddings with clustering methods to discover recurring layout families.
- Validated the pipeline across two independent datasets, producing coherent clusters and interpretable structural representations.

PROJECTS

Fault-Tolerant Distributed Job Processing System

Python, FastAPI, Redis Streams, PostgreSQL

- Built a fault-tolerant asynchronous job processing platform with worker pools, explicit job lifecycle management, retries, dead-letter queues, and persistent state-transition history.
- Implemented at-least-once delivery using Redis consumer groups, stale-message reclamation, exponential-backoff retries, and hash-based idempotency for safe recovery from worker crashes.

Hybrid Search & Retrieval Engine

Python, FastAPI, FAISS, Sentence Transformers

- Built a hybrid information-retrieval engine combining a from-scratch inverted index and BM25 lexical retrieval with dense Sentence Transformer embeddings and FAISS vector search.
- Implemented Reciprocal Rank Fusion and a labeled evaluation pipeline to compare lexical, dense, and hybrid retrieval, with reproducible indexing, API serving, and performance benchmarking.

Distributed Bitcoin Mining Framework

C++, MPI, Pthreads, OpenSSL

- Designed parallel nonce-search implementations using static multithreading, dynamic work stealing, and MPI-based distributed execution with SHA-256 hashing.
- Implemented dynamic work distribution, Raft-based leader election, checkpointing, and fault recovery while analyzing scalability, load balancing, and communication overhead.

RISC-V Processor & Cache Simulator

C++, Computer Architecture

- Built a configurable RV32I processor simulator with instruction decoding, register and memory modules, execution support, and command-line debugging facilities.
- Integrated cache simulation across direct-mapped and set-associative configurations and analyzed hit/miss behavior under different cache organizations and access patterns.

SCHOLASTIC ACHIEVEMENTS

JEE Advanced 2023 AIR 953, Top 0.5%

JEE Mains 2023 AIR 1541, Top 0.15%

COMPETITIVE PROGRAMMING

Codeforces 1400+ rating 

LeetCode 500+ problems solved 

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, JavaScript

Backend: FastAPI, Node.js, Express.js, REST APIs

Systems: Redis, PostgreSQL, MongoDB, MPI, Pthreads

ML/IR: PyTorch, NLP, BM25, FAISS, Sentence Transformers

Tools: Git, GitHub, Linux, CMake, Catch2

Concepts: DSA, OS, DBMS, Networks, Distributed Systems

RELEVANT COURSEWORK

Computer Science: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, Distributed Computing, Computer Architecture, Compiler Design, Software Engineering

Mathematics: Discrete Mathematics, Probability and Random Variables, Calculus